

Numerical Simulation of Learning Dynamics in Multimodal RL

A tunable diversity correction for outcome-level mode collapse

Ahnaf Tahmid (2105041)

Sadia Binte Sayed (2105045)

Mohammad Raihan Rashid (2105046)

Ruwad Naswan (2105051)

Md. Mehedi Hasan (2105052)

CSE 402 Numerical Analysis and Simulation Sessional

Department of Computer Science and Engineering, BUET

Background

Base Paper: The Collapse Mechanism

Sinha, Elango and Liu (2026), *arXiv:2601.21669*.

Many RL tasks have **several equally good answers**: valid reasoning chains, distinct drug candidates, different correct programs. We want the policy to keep all of them, not pick one.

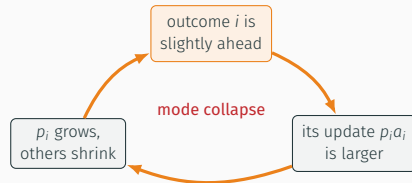
The standard objective is expected return

$$J(\theta) = \mathbb{E}_{o \sim p_\theta} [r(o)] = \sum_i p_i r_i.$$

Under gradient flow the paper proves (Thm. 3.1)

$$\dot{z}_i = p_i a_i, \quad a_i = r_i - \bar{r}.$$

An outcome's update is scaled by **how likely it already is**. That is a rich-get-richer loop, and it runs even when two outcomes have identical reward.



Collapse is a property of **the objective itself**. Entropy bonuses and KL penalties push against it, but they do not remove the cause.

Base Paper: Inverse Probability Scaling

Inverse Probability Scaling. Divide the reward by the probability of the outcome that produced it:

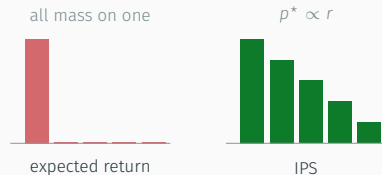
$$\tilde{r}(o) = \frac{r(o)}{p_{\theta}(o)}.$$

The $1/p$ cancels the p_i multiplier, so (Thm. 4.1, Cor. 4.2)

$$\dot{z}_i = r_i - p_i \sum_k r_k \implies p_i^* = \frac{r_i}{\sum_k r_k}.$$

In practice $p_{\theta}(o)$ is unknown. IPS-GRPO estimates it from a group of G sampled rollouts and clips (their Eq. 9):

$$\tilde{r}(o_g) = \frac{r(o_g)}{\max\{\hat{p}(o_g), \varepsilon\}}.$$



Left open by the paper:

1. Only two settings exist: plain (collapse) or exact proportionality.
2. The group size G and clip ε are ablated empirically only (their Table 4).
3. Their own limitation section: finite-group estimates “can be noisy, requiring probability clipping that introduces a bias-variance trade-off”. No analysis is given.

Project Overview

Problem Statement and Extension

The paper gives one fixed correction. We asked three questions it does not answer.

1. Can the correction be tuned?

Replace r/p by r/p^α . Redoing the paper's own gradient calculation gives

$$\dot{z}_i = r_i p_i^{1-\alpha} - p_i \sum_k r_k p_k^{1-\alpha},$$

$$p_i^* \propto r_i^{1/\alpha}$$

One knob spans collapse, the paper's result, and near-uniform.

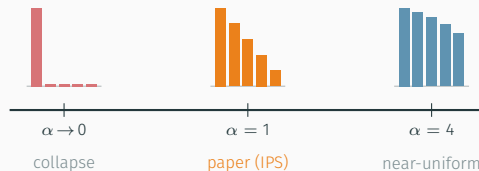
2. What does a finite group cost?

Derive the effect of G and ε instead of ablating it.

3. How wrong is the estimate $\hat{p}$?

The error analysis the paper lists as an open limitation.

stationary policy p^* for $r = (5, 4, 3, 2, 1)$



α becomes a **diversity dial**: the same algorithm, one extra exponent.

System Model

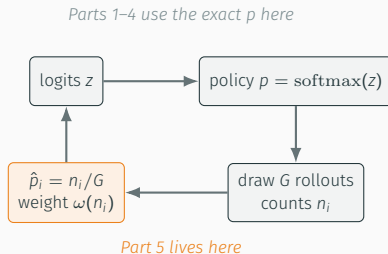
An **outcome-selection bandit**: K outcomes, fixed rewards $r_i > 0$, policy stored as logits and squashed by softmax,

$$p_i = \frac{e^{z_i}}{\sum_k e^{z_k}}.$$

Nothing else is in the model. No network, no exploration schedule, no critic. So any collapse we see has **exactly one possible cause**, which is the point the paper argues.

Two objects are studied throughout:

- the **idealised flow** $\dot{z} = f(z)$, an ODE in K variables;
- the **sampled update**, one noisy step built from G draws.



$$\hat{g}_i = \hat{p}_i r_i \omega(n_i) - p_i \sum_k \hat{p}_k r_k \omega(n_k), \quad z \leftarrow z + h \hat{g}$$

Work Breakdown

Part	Focus	Key content
1	Collapse baseline, $\alpha = 0$	Integrate the flow with Euler and RK4, verify orders, show the exact tie is provably flat, and that sampling noise alone still collapses it
2	The α knob	Stationary law $p^* \propto r^{1/\alpha}$, linearisation and eigenvalue spectrum, convergence rates, explicit-method stability limits, first finite- G ceiling
3	Solving $\dot{z} = 0$ directly	Bisection, secant and Newton on three equivalent formulations; convergence orders, basins, conditioning; nested solver for the finite- G fixed point
4	Multi-parameter atlas	19,125 stationary points over $(\alpha, G, \varepsilon, \rho)$, support and entropy for general K , and whether interpolation can replace solving
5	The estimator itself	Bias and variance of the clipped weight, Richardson extrapolation, golden-section search, power and QR methods, moving averages

One shared library (`src/`), one script per part (`experiments/`), one write-up per part, and 68 unit tests.

Part 1: Collapse Baseline

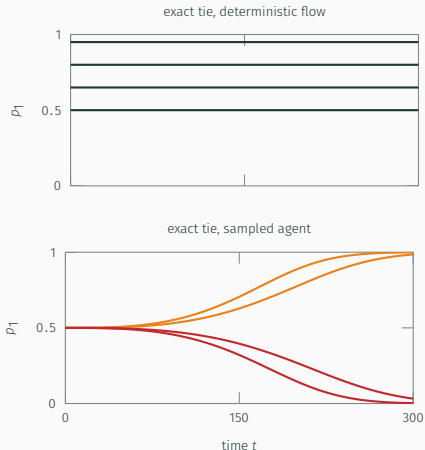
Part 1: Objective

Before correcting collapse we had to reproduce it, and to be precise about a claim the paper states informally.

The paper says collapse happens “even with exactly equal rewards”. For the **deterministic** flow that is not literally true: if all r_i are equal then $a_i \equiv 0$, so $\dot{z} \equiv 0$ and nothing moves at all.

What actually drives collapse under a tie is **sampling noise**. A finite group never sees the exact mean reward, so the realised advantage is never zero, and the p_i multiplier then amplifies it.

So Part 1 runs both and separates them.



Same rewards, same starting policy. Only the noise differs.

Part 1: Numerical Methods

Solution of ODEs. The flow is a coupled system in K unknowns,

$$\dot{z}_i = r_i p_i - p_i \sum_k r_k p_k.$$

Integrated with Euler $z_{n+1} = z_n + hf(z_n)$ and classical RK4

$$z_{n+1} = z_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

Truncation and round-off error. Empirical order from a step-size sweep,

$$\text{order} = \text{slope of } \log E \text{ vs } \log h.$$

Below $h \approx 0.025$ the RK4 error stops falling: the truncation term has dropped under the round-off floor.

Least-squares regression. Collapse time against reward gap, fitted in log-log space,

$$\log t_{\text{col}} = m \log \Delta + c \Rightarrow t_{\text{col}} \approx 53.94 \Delta^{-1.00}.$$

The same fit gives the group-size law $t_{\text{col}} \propto G^{1.07}$.

Random number generation. The sampled agent draws outcomes by **inverse transform sampling**: build the CDF of p , draw $u \sim U(0, 1)$, take

$$o = \min\{i : \sum_{k \leq i} p_k \geq u\}.$$

Visualisation. Semi-log and log-log axes, since both the transition and the fitted laws span decades.

Part 1: Results and Analysis

The tie is provably flat. Four different starting policies, $T = 300$:

$$\max_t |\Delta p_1| = 1.11 \times 10^{-16}$$

which is machine epsilon, not a slow drift. This **sharpens** the paper's claim rather than contradicting it.

Noise alone collapses it. From an exactly symmetric start ($z_0 = 0$), $K = 5$, 200 seeds:

outcome	O1	O2	O3	O4	O5
won	18.5%	22.5%	19.0%	19.0%	21.0%

All inside the 95% band [14.5, 25.5]% around $1/K$. The winner is arbitrary.

Two clean scaling laws.

$$t_{\text{col}} \approx 53.94 \Delta^{-1.00}, \quad R^2 = 0.99999994$$

$$t_{\text{col}} \propto G^{1.07}, \quad R^2 = 0.974$$

A bigger group delays collapse roughly linearly. That is a statement about a real GRPO hyperparameter.

Integrators behave.

	order	error at $h = 0.05$
Euler	1.00	1.4×10^{-4}
RK4	3.99	5.1×10^{-8}

Reported honestly: we looked for an Euler instability here and found none. The right-hand side is bounded and self-damping at $\alpha = 0$, so no step size can overshoot. Part 2 is where instability appears.

Part 2: The Diversity Exponent

Part 2: Objective

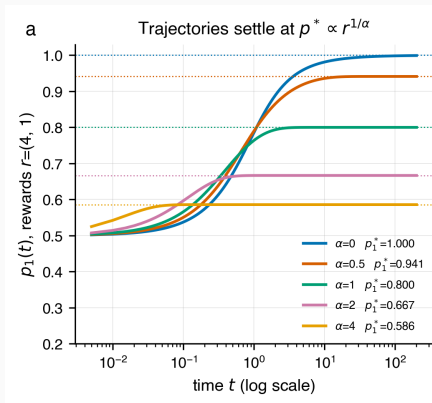
What training does. The model gives each answer a probability. At every step, answers that earn more reward are pushed up. Training ends when nothing changes any more.

Our change. The paper divides each reward by p . We divide by p^α , so α is a **dial**. Training then ends at

$$p_i^* \propto r_i^{1/\alpha}.$$

Our four questions.

1. Does training really end there?
2. How fast does it get there?
3. How big a solver step is safe?
4. What changes with a small sample group?



Two answers with rewards 4 and 1, starting at 50/50. Each curve is one training run: the better answer's share over time. Each α ends at a different share: 100% at $\alpha = 0$, 94% at 0.5, 80% at 1, 67% at 2.

Part 2: Numerical Methods (1/2)

Differential equations. Training is an ODE, $\dot{z} = f(z)$. Setting $\dot{z} = 0$ gives the end point.

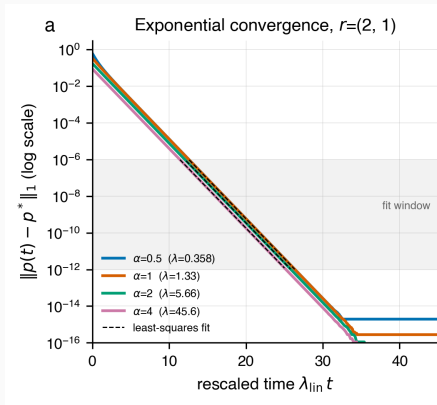
Eigenvalues: the speed. Near the end point the error shrinks like $e^{-\lambda t}$. The speeds λ are the eigenvalues of the Jacobian

$$J^* = -\alpha \|r\|_{1/\alpha} (\text{diag } p^* - p^* p^{*\top}).$$

One eigenvalue is always 0: adding the same number to every score changes nothing.

Least squares: measuring the speed. Fit the slope of the log error.

Round-off: where to fit. Only between 10^{-6} and 10^{-12} , before the precision floor.



Distance from the end point p^* (log scale), rewards 2 and 1. Time is multiplied by each run's *predicted* speed λ (0.36 to 45.6), so if the prediction is right all four lines share one slope. They do. Grey band: fit window. Flat tails: precision floor.

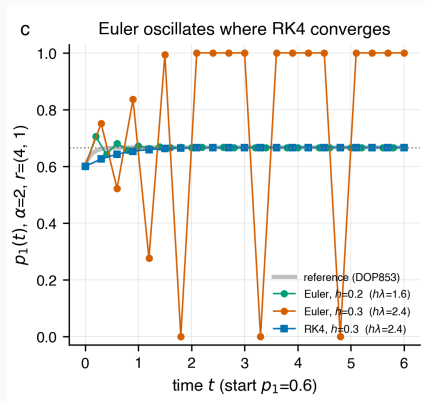
Part 2: Numerical Methods (2/2)

Stability: how big a step? Each step multiplies the error by a factor. Above 1, the error grows and the run blows up. Safe steps:

$$h\lambda_{\max} < 2 \text{ (Euler)}, \quad h\lambda_{\max} < 2.785 \text{ (RK4)}.$$

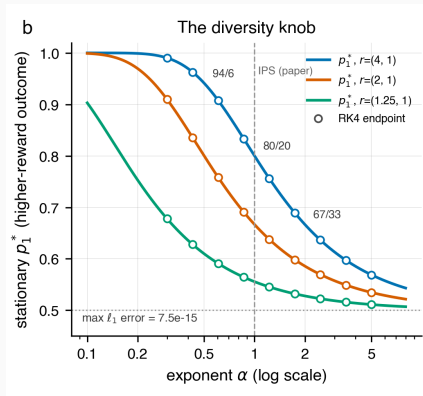
Bisection. 2.785 is the root of $x^3 - 4x^2 + 12x - 24 = 0$, found by bisection. Bisection on the real simulation finds the largest safe step; it matches the theory within 0.03%.

Random sampling. Real training only sees G samples: $G\hat{p}_i \sim \text{Binomial}(G, p_i)$. Averaging over groups gives an **effective weight** $w_G(p)$ in place of $p^{-\alpha}$. The end point solves $r_i w_G(p_i) = S$, by bisection. Simulations are checked against it.

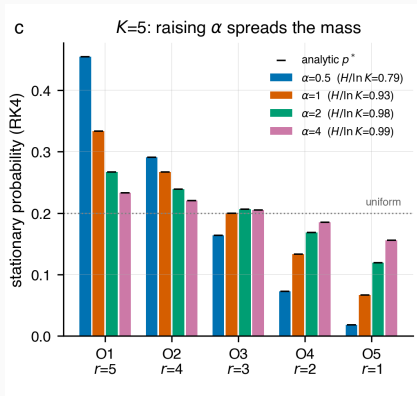


Rewards 4 and 1, $\alpha = 2$, so $\lambda = 8$. Euler with $h = 0.3$ ($h\lambda = 2.4 > 2$) jumps between 0 and 1. RK4 with the same step settles, and so does Euler with $h = 0.2$.

Part 2: Results (1/3): The Dial Works

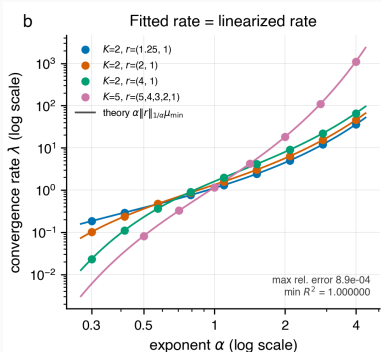


Final share of the better answer against α , for three reward pairs. Circles are simulations, lines are the formula; they agree to 7.5×10^{-15} . “94/6” means the better answer ends with 94% and the worse with 6%.

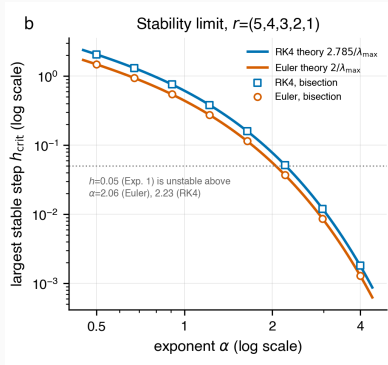


Five answers, rewards 5, 4, 3, 2, 1. A bigger α spreads the probability more evenly. At $\alpha = 1$ it is exactly reward over total reward.

Part 2: Results (2/3): Speed and Step Size



Measured speed (dots) against the eigenvalue prediction (lines): within 0.002% for two answers, 0.09% for five.



Largest safe step: bisection (markers) lands on the theory lines. Part 1's step $h = 0.05$ blows up above $\alpha = 2.06$.

Both ends of the dial are hard. Small α : speeds differ by up to a million times (stiff). At $\alpha = 0$ there is no speed at all: the loser fades like $1/t$.

Part 2: Results (3/3): A Small Group Caps the Boost

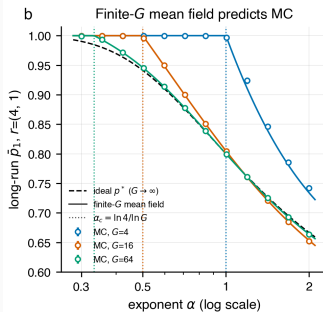
A group of G samples sees a rare answer at most once, so it never estimates p below $1/G$. The boost is capped:

$$\text{boost} \leq \min\{G, 1/\varepsilon\}^\alpha.$$

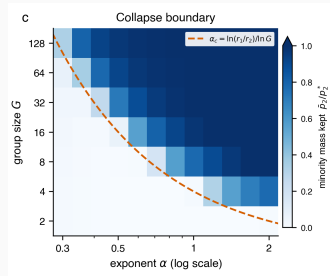
The weaker answer survives only if the cap beats the reward ratio:

$$\alpha > \frac{\ln(r_1/r_2)}{\ln \min\{G, 1/\varepsilon\}}$$

With $G = 4$, the paper's $\alpha = 1$ is not enough.



Rewards 4 and 1. Circles are simulations with groups of 4, 16, 64; lines are our prediction. With $G = 4$ the weaker answer is dead (share 1.0) until α passes 1.



How much of the weaker answer survives, over α and G . The dashed line is our formula $\alpha_c = \ln 4 / \ln G$: it traces the edge.

Part 3: Root Finding Methods

Part 3: Objective

Part 2 reached p^* by integrating for many e-folding times. Part 3 solves $\dot{z} = 0$ **directly** and checks the root against the analytic answer.

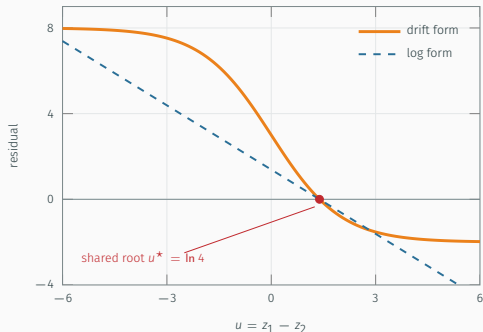
The same condition can be written three ways:

$$\text{drift: } F_i = \dot{z}_i$$

$$\text{balance: } F_i = r_i p_i^{-\alpha} - r_K p_K^{-\alpha}$$

$$\text{log: } F_i = \ln \frac{r_i}{r_K} - \alpha(z_i - z_K)$$

They share the same root. Dividing by p_i removes the very multiplier the paper blames for collapse, and it turns out to matter for Newton too.



The drift saturates at both ends, so a Newton step taken from far out lands even further away. The log form is a straight line, so Newton solves it in one step.

Part 3: Numerical Methods

Root finding. All three from the syllabus, on the same problem:

bisection: halve a sign-change bracket

$$\text{secant: } u_{n+1} = u_n - F_n \frac{u_n - u_{n-1}}{F_n - F_{n-1}}$$

$$\text{Newton: } u_{n+1} = u_n - F(u_n)/F'(u_n)$$

Measured order from the least-squares slope of $\ln e_{n+1}$ against $\ln e_n$.

Systems of equations. For $K > 2$, Newton solves

$$J(y) \delta y = -F(y)$$

by Gaussian elimination each iteration. J is the analytic Jacobian, not a finite difference.

Conditioning, and a gauge. Shifting every logit leaves $\dot{z}$ unchanged, so $J1 = 0$ and the full $K \times K$ system is **singular** (smallest singular value 7.1×10^{-17}). Fixing $z_K = 0$ removes it. At the root

$$J_{\text{balance}} = -\alpha \|r\|_{1/\alpha} I, \quad J_{\log} = -\alpha I,$$

so both have condition number exactly 1.

Nested root finding for finite G . Inner: invert the monotone $w_G(p_i) = S/r_i$ for each outcome. Outer: choose S with $\sum_i p_i(S) = 1$. Both levels are monotone with a guaranteed bracket, so bisection is safe and safeguarded Newton is fast.

Part 3: Results and Analysis

Textbook orders, confirmed. $K = 2$, $\alpha = 1$, root $u^* = \ln 4$:

method	iterations	order
bisection	56	linear (0.502)
secant	9	1.61
Newton (drift)	5	2.00
Newton (log)	1	exact

How you write the equation decides the basin.

Newton on the drift form converges from only 17% of starts at $\alpha = 1$, but from 100% once $\alpha > 1.113$. On the balance form it converges from 100% of starts at every α .

Backtracking does not rescue it: for $\alpha < 1$ every simplex vertex is a zero of the drift “at infinity”, and a line search is pulled into them.

Nested solver: Newton pays off. Over

$G \in \{4, 16, 64\} \times \alpha \in \{0.5, 1, 2\}$:

	outer iters	total work
bisection / bisection	43–55	4,506–13,769
Newton / Newton	6–16	225–1,703

Same answer to 9.8×10^{-14} , with 8 to 20 times less work.

A new result for $K > 2$. The two-outcome threshold understates the requirement. At $G = 16$,

$r = (5, 4, 3, 2, 1)$:

α_c for	02	03	04	05
$K = 5$ mean field	0.0805	0.203	0.419	0.886
pairwise formula	0.0805	0.184	0.330	0.580

Some outcomes are lost at every α : here the weakest needs $G \geq 6$ before any exponent can keep it.

Part 4: Parameter Atlas

Part 4: Objective

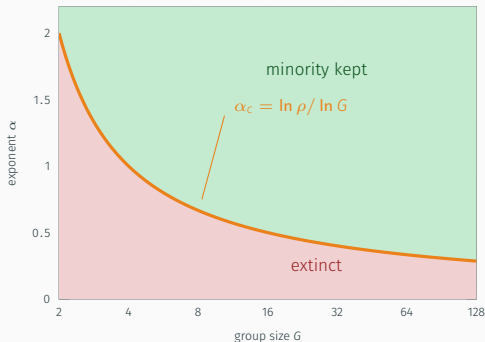
Parts 2 and 3 varied one or two parameters. Part 4 asks whether the survival boundary really holds **jointly** over all four, and whether a grid can stand in for solving.

The predicted boundary collapses to a single signed margin

$$m = \alpha \log \min\{G, \frac{1}{\varepsilon}\} - \log \rho,$$

with $\rho = r_1/r_2$. Survival should be exactly $m > 0$, independent of where in the four-dimensional box you are.

19,125 stationary points were solved to test that one inequality.



One slice ($\rho = 4$, $\varepsilon = 10^{-3}$) of a four-dimensional atlas.

Part 4: Numerical Methods

Interpolation. The atlas is a tensor-product grid in $(\log \alpha, \log \rho, \log \varepsilon)$. Inside one cell, with $t_d \in [0, 1]$ the fractional position along axis d , the **multilinear** interpolant is the weighted sum over the cell's 2^D corners:

$$\hat{f}(\mathbf{x}) = \sum_{\mathbf{c} \in \{0,1\}^D} f(\mathbf{x}_{\mathbf{c}}) \prod_{d=1}^D t_d^{c_d} (1 - t_d)^{1-c_d}.$$

This is the D -dimensional form of the linear interpolation in the syllabus, and it is exact at nodes and for any function affine in each coordinate separately.

Approximation error and order. Refine the grid and fit the slope of $\log(\text{RMSE})$ against $\log(\text{intervals per axis})$.

Root finding, vectorised. All 19,125 $K = 2$ solves run as one batched bisection with 64 deterministic halvings, and 40 of them are re-checked against the original scalar solver.

Model validation and verification. The script refuses to write a summary unless every check passes: phase classification, stationarity residual, normalisation, node exactness, and monotone refinement.

Where the theory says interpolation must degrade. The response is continuous but has derivative kinks at

$$m = 0 \text{ (extinction)}, \quad \varepsilon = 1/G \text{ (clip activates)},$$

so cells crossing those surfaces cannot keep second-order behaviour. We measure the two regions separately instead of quoting one rate.

Part 4: Results and Analysis

The boundary holds everywhere. Over $51 \times 15 \times 5 \times 5 = 19,125$ points:

$$\text{misclassifications} = 0/19,125$$

with worst stationarity residual 1.18×10^{-15} .

Two separate ceilings. ε can bind before G does. At $\alpha = 1, \rho = 4$:

p_2 at	$G=6$	$G=16$	$G=64$	$G=256$
$\varepsilon = 10^{-3}$	0.115	0.195	0.200	0.200
$\varepsilon = 0.3$	0	0	0	0

With $\varepsilon = 0.3$ the clip caps the boost at $3.33 < 4$, so **no group size whatsoever** rescues the minority.

Group size must grow with the problem. Smallest G that keeps every outcome at $\alpha = 1$:

K	spread 1.25	spread 32
2	2	33
5	3	54
10	4	96

A best-versus-worst ratio alone does not predict this once several strong outcomes compete for mass.

Interpolation audit. 800 held-out off-grid points:

region	RMSE (17^3)	order
smooth	2.5×10^{-3}	1.58
near a kink	9.7×10^{-3}	1.03

So a grid is fine for exploring, but survival decisions near $m = 0$ need a real solve. The margin m is useful twice: as the criterion, and as a warning flag.

Part 5: Estimator Error Analysis

Part 5: Objective

The problem. IPS needs the weight $p^{-\alpha}$ for every answer, but training never knows p . It draws a group of G samples, counts how often each answer appears (n), and uses the estimate

$$\omega(n) = \max \left\{ \frac{n}{G}, \varepsilon \right\}^{-\alpha} \approx p^{-\alpha}.$$

The clip ε stops a tiny count from giving a huge weight.

The paper says this estimate “can be noisy, requiring probability clipping that introduces a bias-variance trade-off”, but it never measures that trade-off. Part 5 does, in three steps.

Goal 1: Measure the error.

How wrong is $\omega(n)$, and what is the best clip ε ?



Goal 2: Try to do better.

Three other ways to estimate the weight.



Goal 3: What training feels.

Which of these errors change where training ends up?

Part 5: Numerical Methods (1/3): Measure the Error

1a. How big is the error?

Taylor expansion

The weight is $f(\hat{p})$ with $f(x) = x^{-\alpha}$ and $\hat{p} = n/G$.
Expand around the true p , treating $\hat{p} - p$ as a random step:

$$f(\hat{p}) \approx f(p) + f'(p)(\hat{p} - p) + \frac{1}{2}f''(p)(\hat{p} - p)^2.$$

Average both sides. The middle term vanishes because $\mathbb{E}[\hat{p}] = p$, and $\mathbb{E}[(\hat{p} - p)^2] = \frac{p(1-p)}{G}$. This gives the relative bias and spread:

$$b \approx \frac{\alpha(\alpha+1)(1-p)}{2Gp}, \quad s \approx \alpha \sqrt{\frac{1-p}{Gp}}.$$

The bias comes from the curve of $x^{-\alpha}$ (the f'' term).
Both shrink with Gp , the expected number of times the answer is seen.

1b. What is the best clip?

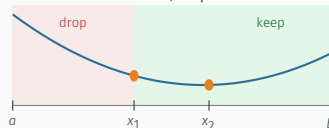
Golden-section search

Total error: $\text{MSE}(\varepsilon) = b^2 + s^2$.

- Small ε : rare answers get huge weights (noise).
- Large ε : many weights are capped too low (bias).

So the error falls, then rises. It has a kink at every $\varepsilon = n/G$, so it has no derivative there and Newton's method does not fit.

Golden-section search only compares values. Keep two test points at ratio 0.618, drop the side that cannot hold the minimum, repeat:



Part 5: Numerical Methods (2/3): Try Better Estimators

Split-half Richardson.

Richardson extrapolation

From Goal 1, the average weight from a group of G has the form

$$E_G = f(p) + \frac{c_1}{G} + \frac{c_2}{G^2} + \dots$$

with c_1, c_2 not depending on G . A group half the size gives $E_{G/2} = f(p) + \frac{2c_1}{G} + \frac{4c_2}{G^2} + \dots$. Combine the two so the $1/G$ term cancels:

$$2E_G - E_{G/2} = f(p) - \frac{2c_2}{G^2} + \dots$$

No extra samples are needed. Split the group into halves A, B : each is a real group of $G/2$.

$$\omega_R = 2f(\hat{p}) - \frac{1}{2} [f(\hat{p}_A) + f(\hat{p}_B)]$$

Guard: use ω_R only if both halves saw the answer and $\omega_R > 0$. Otherwise use the plain clip.

Laplace smoothing. Pretend every answer was seen λ extra times:

$$\omega_\lambda(n) = \left(\frac{n + \lambda}{G + K\lambda} \right)^{-\alpha}.$$

The estimate is never 0, so there are no huge weights.

Moving average over steps. Reuse past groups instead of one:

$$\bar{p}_t = (1 - \beta) \bar{p}_{t-1} + \beta \hat{p}_t.$$

Less noise, for free. But $\bar{p}$ lags behind a policy that is still changing. **Golden-section search** picks the best β .

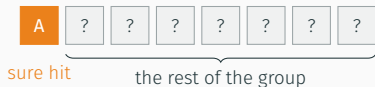
For each rule we measure its bias and spread, as in Goal 1.

Part 5: Numerical Methods (3/3): What Training Feels

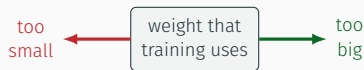
Key idea: training only learns from answers it actually saw.

If an answer is missing from the group, its weight is multiplied by zero. It is never used, however wrong it is.

So the weight that matters is the one from a group that **already contains the answer**: one sure hit, plus the rest of the group.



Taylor expansion again, on this kind of group. It finds two small errors that pull in **opposite directions**:

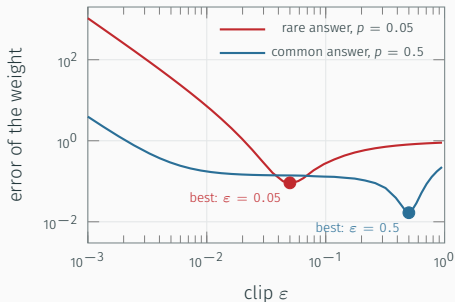


- **The sure hit** makes the answer look a little more common, so its weight comes out a little **too small**.
- **The curve** of the weight formula (as in Goal 1) makes it come out a little **too big**.

Add the two: that is the error training feels. If it is zero, training ends exactly at the ideal share.

Check: train with each rule and measure how far training ends from the ideal share.

Part 5: Results (1/2): Measuring and Improving the Estimate



Error of the weight against the clip ($G = 16$). Each answer is best at $\epsilon = \text{its own } p$. The two best points are far apart.

Goal 1: measuring the error

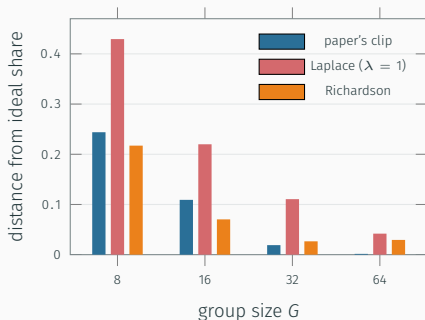
- Error shrinks the more often the answer is seen.
- Best clip: **exactly $\epsilon = p$** (proved; golden section agrees).
- One ϵ for all answers: **no single clip suits common and rare answers.**

Goal 2: three other estimators

- **Richardson:** error shrinks much faster as G grows.
- **Moving average:** less noise for free, but it lags.
- **Laplace:** less noise, but a weaker boost for rare answers.

On paper, Richardson and the moving average are **better estimators.**

Part 5: Results (2/2): What Training Feels



Real training: 5 answers, 32 runs each. Lower means training ended closer to the ideal share. At $G = 64$: clip 0.0016, Richardson 0.029.

Goal 3: what training feels

- At the paper's own setting ($\alpha = 1$), the two errors are **exactly the same size**, so they cancel. The paper's simple clip is almost perfect for training.
- Richardson removes one of the two errors, so the other one is left alone. It wins with small groups but **loses with large ones**.
- Laplace is the worst at every group size.

Main lesson. A better estimate does not always mean better training. What matters is the error that **training actually feels**.

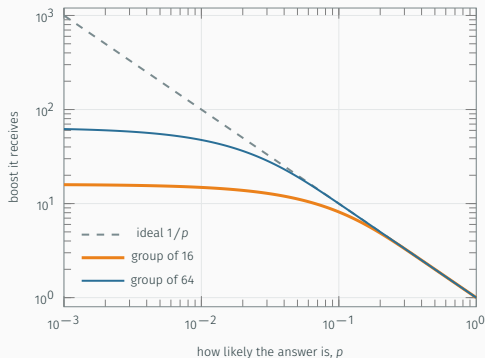
Overall Findings

Comparative Analysis

Question	The paper	What we added
Does expected return collapse onto one answer?	Proved it	Made it precise: with an exact tie nothing moves at all (10^{-16}). The random sampling is what picks a winner.
Does IPS give share $\propto$ reward?	Proved it	Confirmed to machine precision, and extended to any dial setting: share $\propto r^{1/\alpha}$.
Do the group size G and clip ϵ matter?	Tried a few values (their Table 4)	Derived the exact rule: the weaker answer survives only if $\alpha > \ln \rho / \ln \min\{G, 1/\epsilon\}$. Checked on 19,125 settings with no exceptions.
Does clipping cause a bias-variance trade-off?	Named it as a limitation	Measured it exactly. The best clip is $\epsilon = \rho$, and we compared three other rules.

We did not rerun the paper's language-model or molecule experiments; they need far more computing than this course. We reproduced the *mechanism*, in a small model where nothing else can explain the result.

Cross-Cutting Result: The Finite-Group Ceiling



One result that every part runs into.

The ideal boost $1/p$ keeps growing as an answer gets rarer (dashed). A real group of G samples can only give a limited boost (solid lines), because a rare answer shows up in the group at most once:

$$\text{boost} \leq \min\{G, 1/\varepsilon\}^\alpha.$$

Each part meets this cap from a different side:

- Part 2 finds the cap and the two-answer rule
- Part 3 with more answers, a bigger α is needed
- Part 4 checks the rule on 19,125 settings
- Part 5 only the size of the cap decides survival

At $\alpha = 1$ the solid curves are exactly $(1 - (1 - p)^G)/p$ (Part 5).

Conclusions and Limitations

Advice for anyone using this

- **Choose the group size first.** If $\min\{G, 1/\varepsilon\}$ is smaller than best reward $\div$ worst reward, no α can keep the weak answers.
- **Keep ε at or below $1/G$.** A bigger clip adds a second cap.
- **With many answers, the two-answer rule is too hopeful.** At $G = 16$ the weakest of five needs $\alpha = 0.886$, not 0.580.
- **Averaging over steps is cheap noise reduction,** like a $19\times$ bigger group at $\beta = 0.1$. With too little damping ($\beta < 4h\lambda$) training oscillates.

What we do not claim

- We studied a simple model: one choice among K answers. It is not full GRPO with its reward normalisation and PPO clipping, so our results are about the *weight rule*.
- Our α -matched correction is accurate to second order, not exact.
- Averaging per answer needs a short list of possible answers. That is true here, not for generated text.
- We show convergence in simulations from many starting points; we do not prove it for every α .

Thank you

Questions